

Exploration of database

Statistics about sparsity of data contained in the ArXivDigest database.

Classification of signals

- **User self-provided information:** user-inserted information from table `feedback` :
`feedback_text` , `feedback_values` .
- **Authorship-based signals:** published papers authored by users collected from APIs.
- **Historical signals:** clicks and interactions such as `read` , `seen` , `saved` , `seen_website` ,
`seen_email` .
- **Time-based information:** dates of recommendations and last recommendation date; idea:
recommendations with similar scores/clicks can be narrowed by recency to reflect more current
user interests.

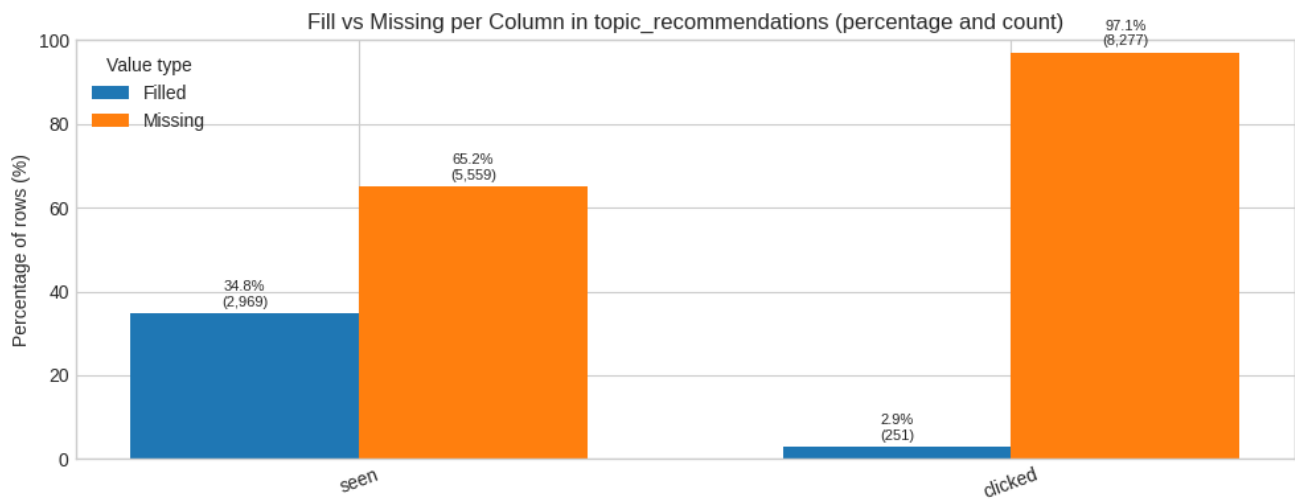
DB test result: 1

```
Engine(mysql+pymysql://maciek:***@localhost:3306/arxiv_db)
```

	profile_type	status	user_count
0	dblp	Linked	49
1	dblp	Not linked	550
2	semantic_scholar	Linked	50
3	semantic_scholar	Not linked	549
4	google_scholar	Linked	100

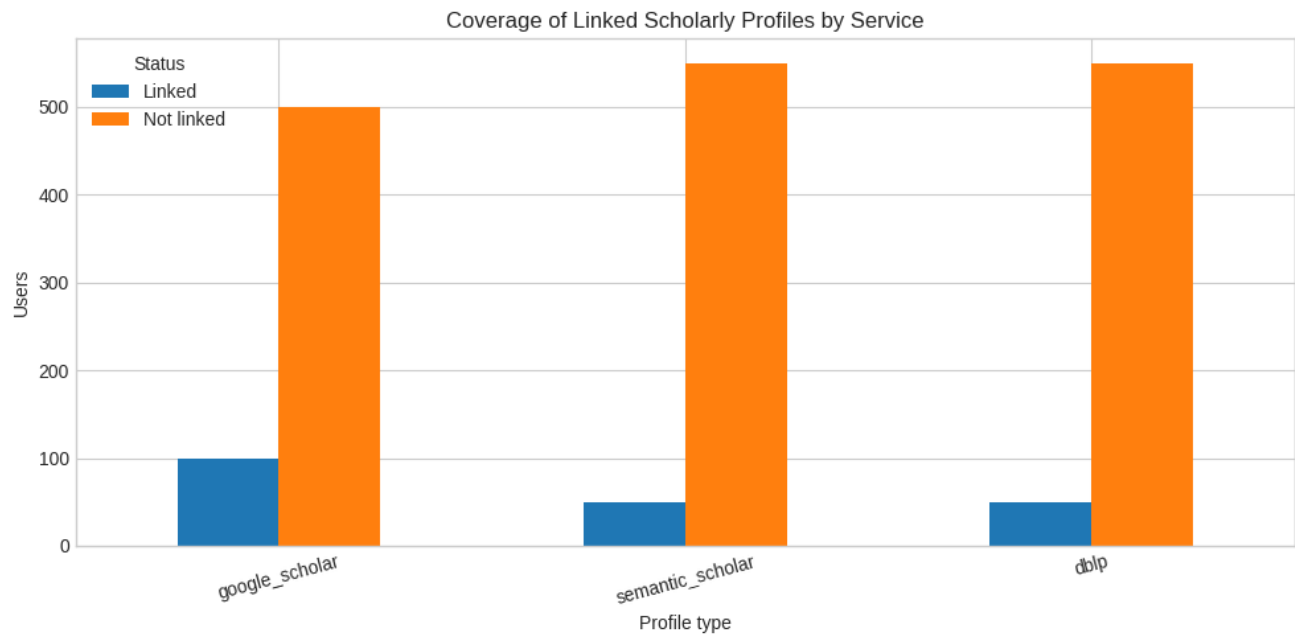
	column_name	non_empty_rows	total_rows	fill_percentage
0	seen	2969	8528	34.81473
1	clicked	251	8528	2.94325

'topic_recommendations_fill_missing_percentage_bar.png'



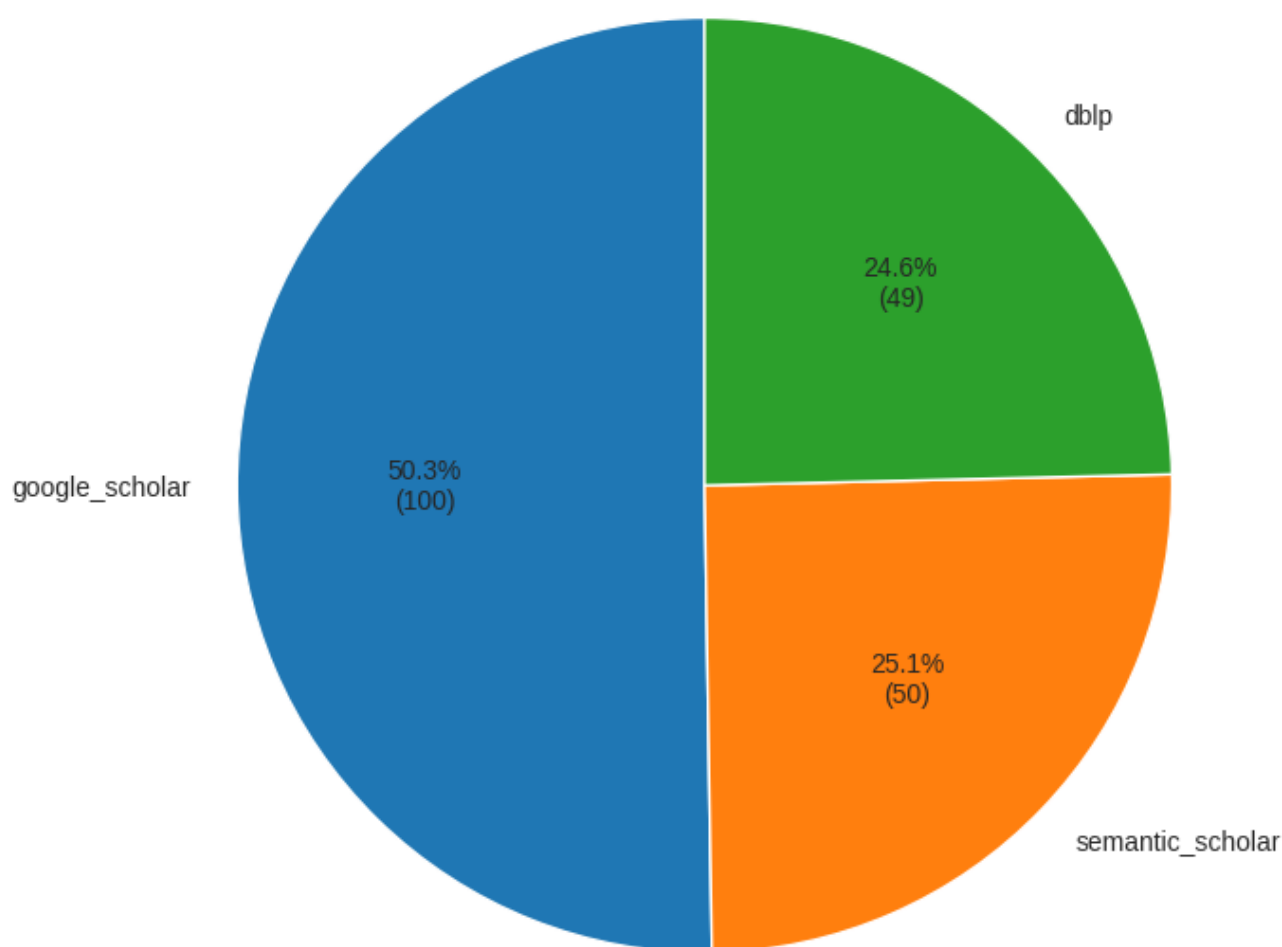
	profile_type	status	user_count
4	google_scholar	Linked	100
5	google_scholar	Not linked	499
2	semantic_scholar	Linked	50
3	semantic_scholar	Not linked	549
0	dblp	Linked	49
1	dblp	Not linked	550

'db_servicesproportions_bar_by_service.png'



'db_servicesproportions_linked_pie_by_service.png'

Share of Linked Profiles by Service



'article_feedback_fill_missing_percentage_bar.png'

Fill vs Missing per Column in article_feedback (percentage and count)

